

IOT104TC PROJECT

Smart Classroom

Energy-Saving, Safety & Asset Monitoring System

Local-first classroom control with real sensors, actuators, MQTT operations, dashboard analytics and AI-assisted commands.

UNO Edge Node

Sensors, state machine
and actuators

UART + ACK

ESP32 Gateway

Wi-Fi, MQTT and
dashboard bridge

9+

hardware signals

5

priority modes

1s

telemetry cadence

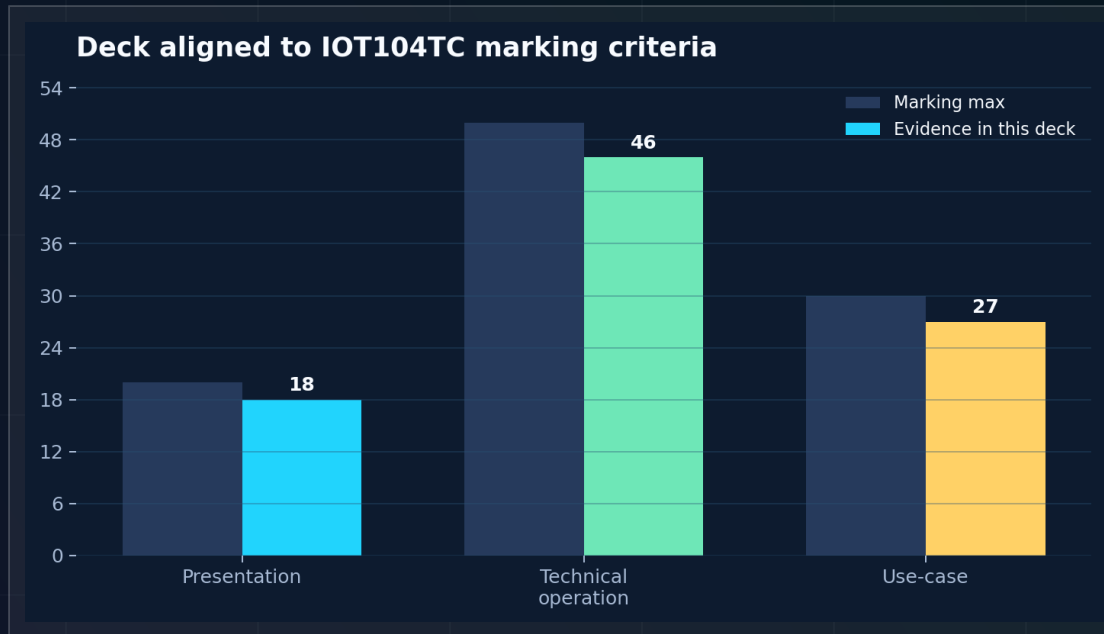
432m

recorded evidence

Local safety works even when the network is offline.

Built around the marking scheme

The demo connects hardware, software, architecture, security/privacy, operations and a practical classroom use-case.



Application

Energy, comfort and safety in one classroom service.

Architecture

UNO local control.
ESP32 MQTT gateway.

Evidence

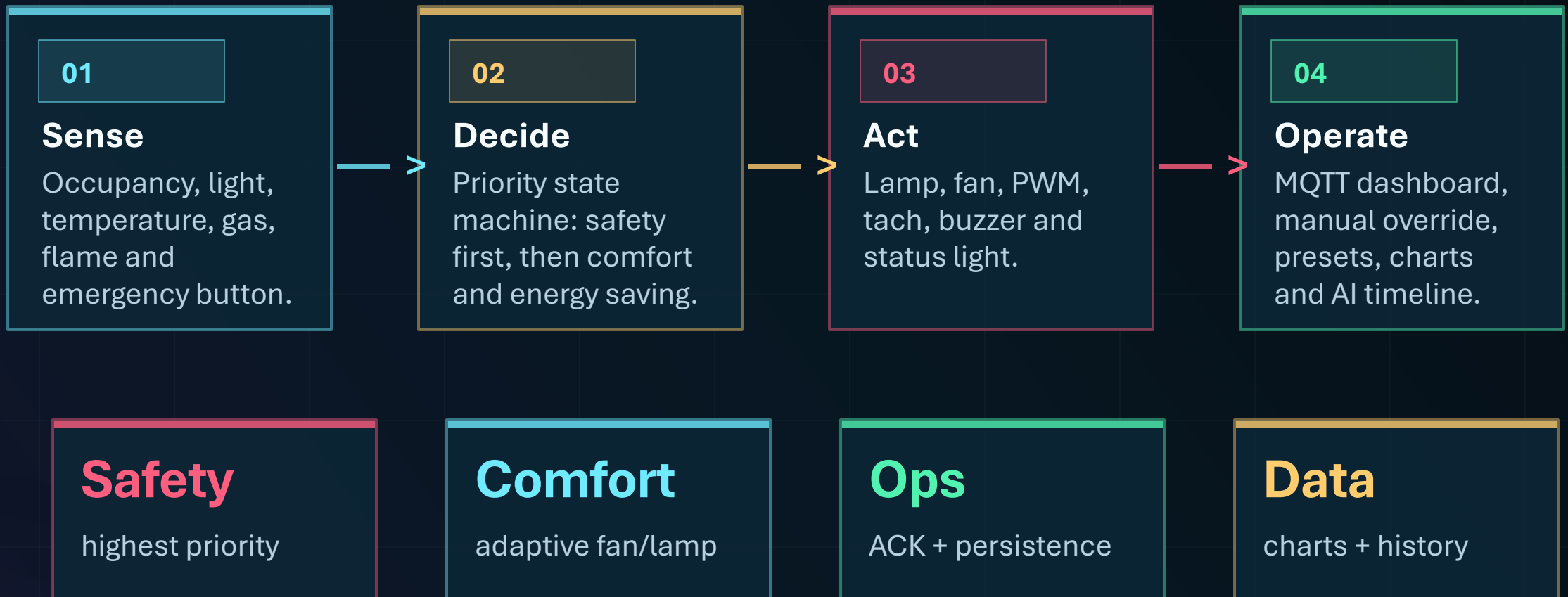
Telemetry, screenshots and self-check.

Security

Fail-safe local control and ignored secrets.

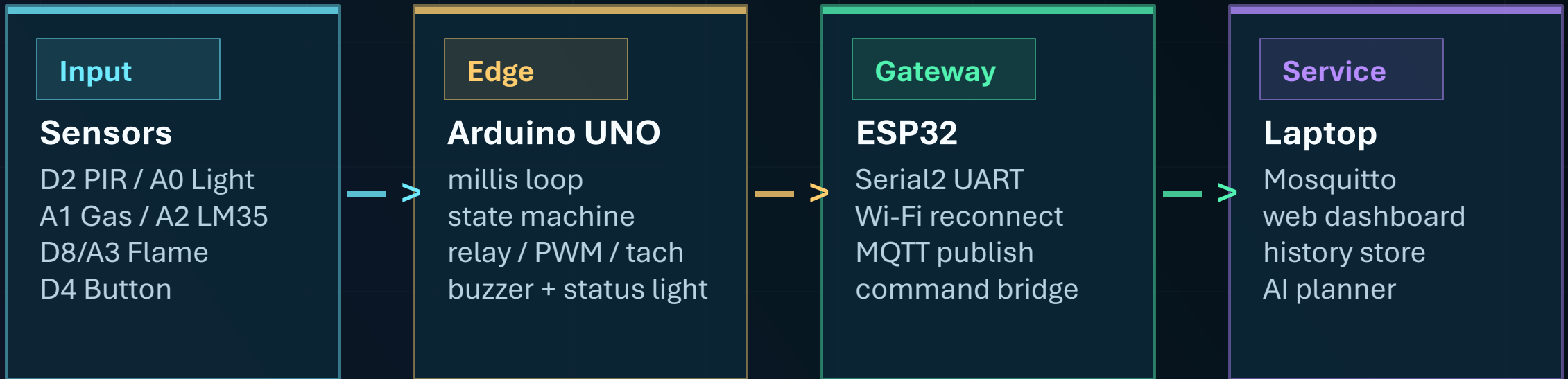
A classroom that reacts locally and reports globally

The system behaves like a small building-management service, not a sensor list.



Real-time control stays on UNO

Networking adds visibility and commands, but does not become a dependency for safety.



Design decision: ESP32 is a gateway; UNO remains the controller of record.

Dense hardware, clean responsibility

The build is complex, but each module has a clear role in the service.

48-kit modules

- #18 PIR
- #31 TEMT6000
- #25 Gas Sensor
- #16 Flame Sensor
- #5 Active Buzzer
- Traffic Light Module

External components

- UNO R3
- ESP32 Gateway
- LM35 powered from ESP32 5V
- 6-channel relay
- 12V 4-wire fan
- LED + resistor

Wiring rules

Common ground is required.
UNO A5 to ESP32 RX2 uses divider.
Relay logic uses RELAY_ON / RELAY_OFF.
No capacitor is added in this build.

ENGINEERING FIX

LM35 became stable after moving power to ESP32 5V and adding ADC guard compensation.

Priority is a state machine

Mixed behavior stays predictable because safety, sensor health, manual control, cooling, lighting and saving are ordered.

SAFETY_ALARM	gas/flame/button
SENSOR_ERROR	invalid sensor state
MANUAL	operator override
COOLING	temperature high
LIGHTING	occupied + dark
ENERGY_SAVING	empty or bright
NORMAL	monitoring

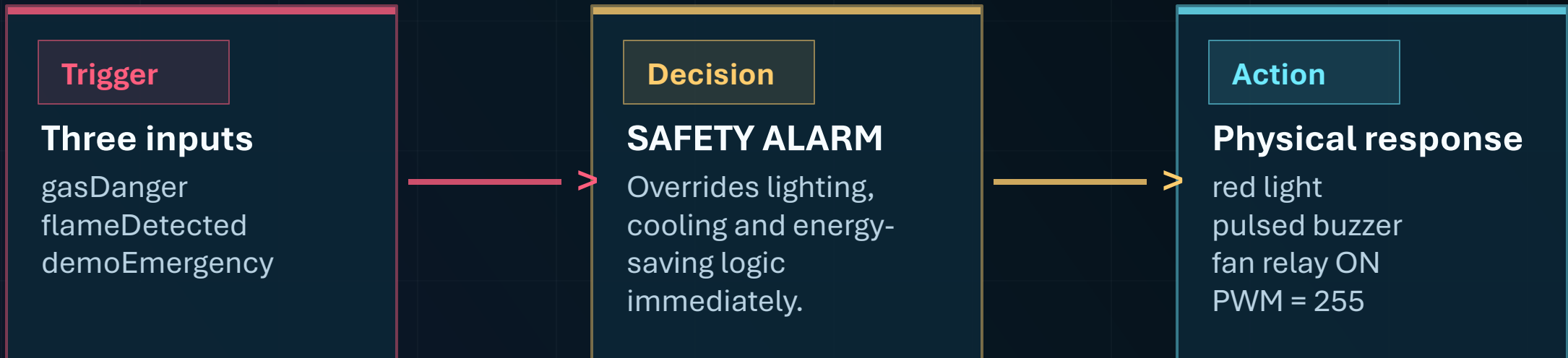
WHY IT MATTERS

Dashboard explains what happened and why.

That turns raw telemetry into an operational classroom service.

The highest-priority path is visible and local

The button gives a safe live demo while gas and flame remain the real sensor paths.



Demo first: press D4, then watch red light, buzzer, fan and dashboard alarm.

Human-friendly automation

The classroom avoids waste without making the controls feel abrupt.

Lighting

Bright room: lamp stays off even if PIR sees motion.

Dark room + occupied: lamp turns on.

No occupancy timeout: lamp turns off.

Cooling

Temperature threshold starts the relay.

PWM ramps fan intensity.

Tach RPM is interrupt-based and never blocks the system.

10s

occupancy timeout

PWM

smooth fan path

ACK

config confirmed

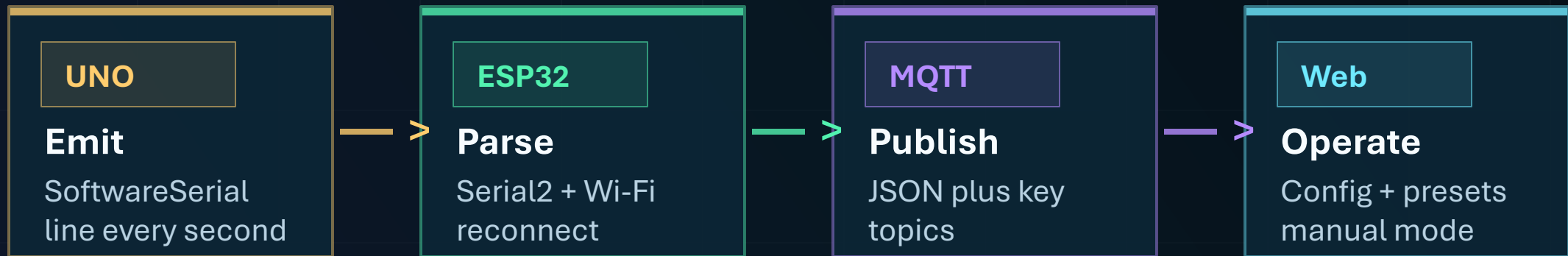
Local

works offline

Telemetry goes up; commands come back with ACK

The protocol is simple enough to debug and structured enough for a dashboard.

```
SC2,mode=SAFETY_ALARM,pir=1,temp10=286,gas=520  
fan=1,pwm=255,rpm=1320,status=R
```



ACK loop proves the command actually landed on the UNO.

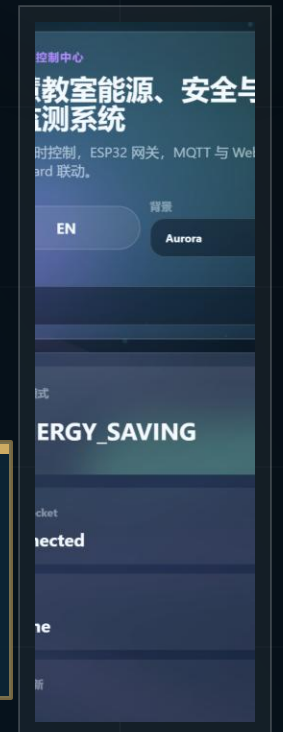
An operator console, not a raw JSON page

Large cards, live charts, mobile access, manual mode and scene presets make the system usable.



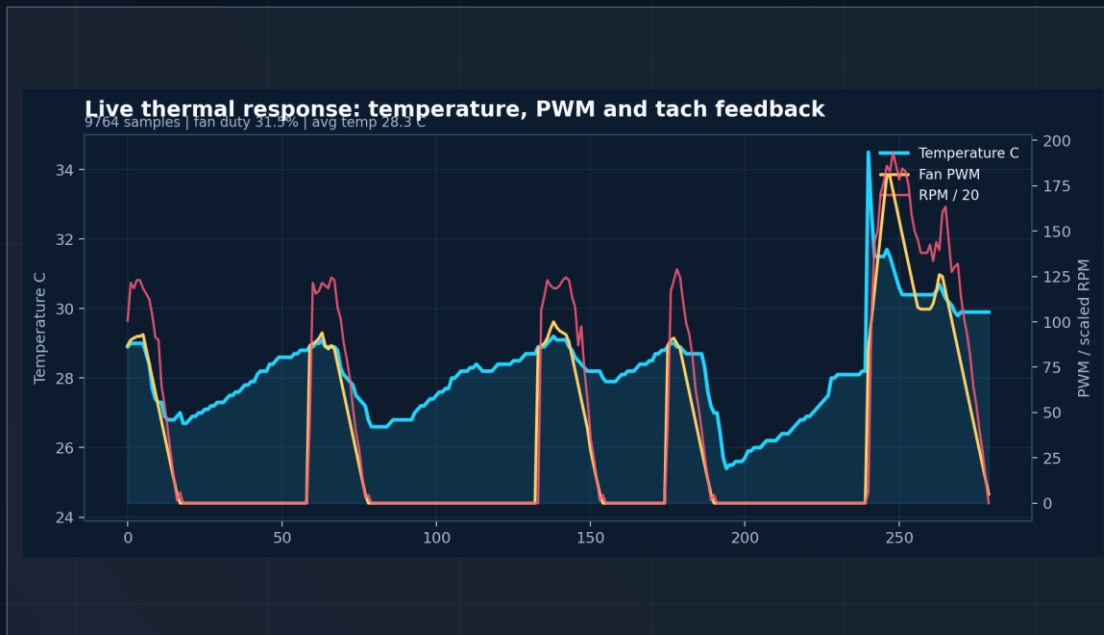
Auto
manual
switch

Scenes
one-click
presets



The demo has history, not just live values

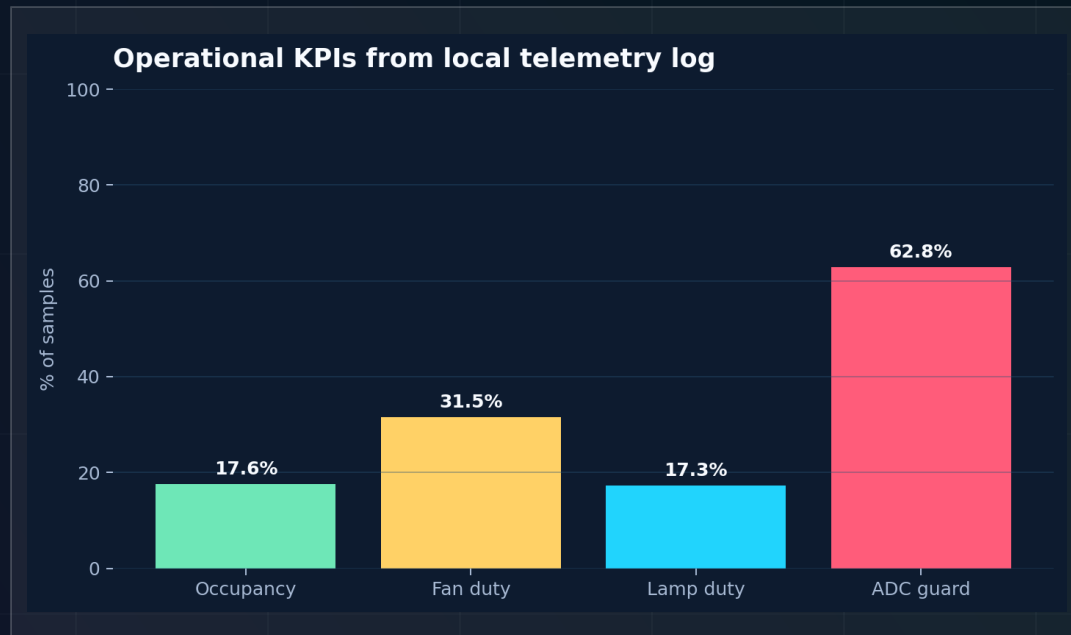
Recorded data supports behavior explanation: modes, temperature response, fan duty and safety events.



9,764 samples | avg temp 28.3C | fan duty 31.5% | safety events 5

Complexity comes from closed loops

Sensors, control, gateway, dashboard, persistence and AI all interact with the same physical system.

**9,764**

telemetry samples

12+

MQTT topics

2-wayUART command +
ACK**Local-first**

safety path offline

34/34delivery self-check
PASS

Natural language becomes structured control

AI is useful because it turns vague operator intent into ordered tasks and timed actions.



Voice

Capture

Browser mic
input

STT

Text

faster-
whisper

Qwen

Plan

structured
tasks

Timeline

Delayed actions auto-clear after
completion.

Data

Overrides
guide comfort.

Demo-friendly, but risk-aware

The project separates safety-critical local control from network convenience and AI assistance.

Safety boundary

Safety alarm, buzzer, red light and fan response are local on UNO.

Privacy boundary

Microphone is user-triggered and requires browser permission.

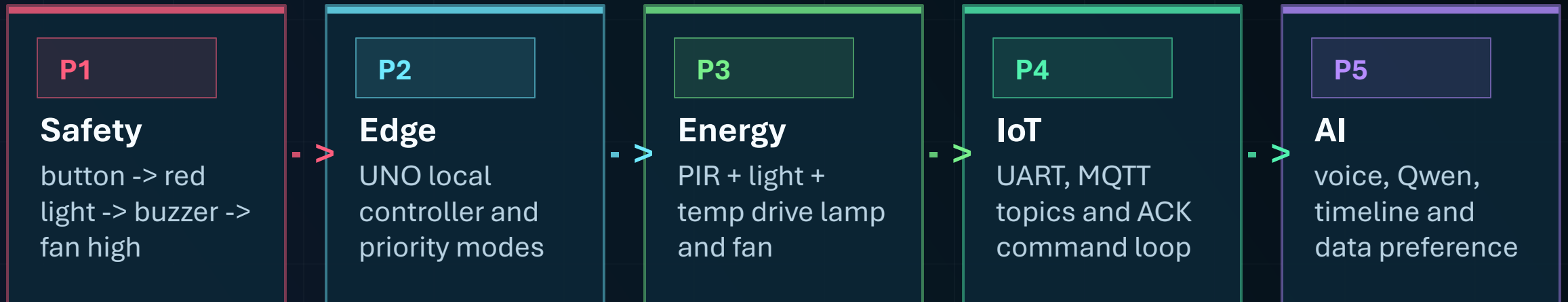
Operations boundary

MQTT status, ACK logs, persistence and self-check reduce demo risk.

Production hardening path: authenticated MQTT, TLS, enclosure, fuse, isolation and power filtering.

Show the strongest engineering first

The audience should see safety and edge control before dashboard polish and AI features.



Do not open with AI. Open with physical safety response.

Ready to inspect, rebuild and present

The final package includes sources, assembly materials, scripts and a repeatable delivery check.

References

- Arduino UNO R3 docs
- Espressif Arduino-ESP32 Serial docs
- Eclipse Mosquitto config docs
- Arduino PubSubClient docs
- Texas Instruments LM35 page

Self-check gates

- PPT structure
- required docs
- firmware presence
- secret scan
- Node/Python syntax
- UNO + ESP32 compile

Final claim: a local-first IoT classroom controller with safety, energy saving, operations and evidence-backed intelligence.